

DNA sequences

Legend:

RED = Met1 of CDS

YELLOW = TEV protease recognition motif

GREEN = Met1 of ζ variants

BLUE = stop codon

UNDERLINED = prolonged sequence at the protein N-terminus (ACKGSSGG)

>14-3-3 ζ _C25A_C189A_(ζ)

ATGGGCAGCAGCCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
GCCATATG**GAAACCTGTATTTCCAGAGT**GTCGAC**ATG**GATAAAAACGAACTGGTGCAGAAAGCGAA
ACTGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCAATGAAATCTGTGACCGAACAGGGC
GCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCTACAAAAATGTGGTTGGTGCACGTC
GCAGCTCTTGGCGTGTCGTGAGTTCCATTGAACAGAAAACGGAAGGTGCTGAGAAAAAACAGCAAAT
GGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGATATCTGTAACGACGTCCTGAGCCTG
CTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTAAAGTTTTCTATCTGAAAATGAAAG
GCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGACAAAAAAGGTATCGTCGATCAGAG
CCAGCAAGCTTACCAAGAAGCGTTTCGAAATCTCTAAAAAAGAAATGCAGCCGACGCATCCGATCCGT
CTGGGTCTGGCCCTGAACCTCAGTGTTTTCTATTACGAAATCCTGAACAGTCCGGAAAAAGCGGCAT
CCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGACACGCTGTCAGAAGAATCGTACAA
AGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACCCTGTGGACGTCCGATACCCAAGGC
GACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT**TAA**

>14-3-3 ζ _C25A_C189A_L12E_M78K_(ζ m)

ATGGGCAGCAGCCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
GCCATATG**GAAACCTGTATTTCCAGAGT**GTCGAC**ATG**GATAAAAACGAACTGGTGCAGAAAGCGAA
AGAGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCAATGAAATCTGTGACCGAACAGGGC
GCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCTACAAAAATGTGGTTGGTGCACGTC
GCAGCTCTTGGCGTGTCGTGAGTTCCATTGAACAGAAAACGGAAGGTGCTGAGAAAAAACAGCAAAA
GGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGATATCTGTAACGACGTCCTGAGCCTG
CTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTAAAGTTTTCTATCTGAAAATGAAAG
GCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGACAAAAAAGGTATCGTCGATCAGAG
CCAGCAAGCTTACCAAGAAGCGTTTCGAAATCTCTAAAAAAGAAATGCAGCCGACGCATCCGATCCGT
CTGGGTCTGGCCCTGAACCTCAGTGTTTTCTATTACGAAATCCTGAACAGTCCGGAAAAAGCGGCAT
CCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGACACGCTGTCAGAAGAATCGTACAA
AGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACCCTGTGGACGTCCGATACCCAAGGC
GACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT**TAA**

>14-3-3 ζ _C25A_C189A_S58D_(ζ _S58D)

ATGGGCAGCAGCCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
GCCATATG**GAAACCTGTATTTCCAGAGT**GTCGAC**ATG**GATAAAAACGAACTGGTGCAGAAAGCGAA
ACTGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCAATGAAATCTGTGACCGAACAGGGC
GCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCTACAAAAATGTGGTTGGTGCACGTC
GCAGCGATTGGCGTGTCGTGAGTTCCATTGAACAGAAAACGGAAGGTGCTGAGAAAAAACAGCAAAT
GGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGATATCTGTAACGACGTCCTGAGCCTG

CTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTAAAGTTTTCTATCTGAAAATGAAAG
 GCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGACAAAAAAGGTATCGTCGATCAGAG
 CCAGCAAGCTTACCAAGAAGCGTTCGAAATCTCTAAAAAAGAAATGCAGCCGACGCATCCGATCCGT
 CTGGGTCTGGCCCTGAACCTCAGTGTTTTCTATTACGAAATCCTGAACAGTCCGGAAAAAGCGGCAT
 CCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGACACGCTGTCAGAAGAATCGTACAA
 AGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACCCTGTGGACGTCCGATACCCAAGGC
 GACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT**TAA**

>14-3-3ζ_C25A_C189A_S57D_S58D_(ζ_S57D_S58D)

ATGGGCAGCAGCCATCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
 GCCATATG**GAAAACCTGTATTTCCAGAGT**GTCGAC**ATG**GATAAAAACGAACTGGTGCAGAAAGCGAA
 ACTGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCAATGAAATCTGTGACCGAACAGGGC
 GCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCTACAAAAATGTGGTTGGTGCACGTC
 GCGACGATTGGCGTGTCGTGAGTTCCATTGAACAGAAAAACGGAAGGTGCTGAGAAAAAACAGCAAAT
 GGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGATATCTGTAACGACGTCCTGAGCCTG
 CTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTAAAGTTTTCTATCTGAAAATGAAAG
 GCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGACAAAAAAGGTATCGTCGATCAGAG
 CCAGCAAGCTTACCAAGAAGCGTTCGAAATCTCTAAAAAAGAAATGCAGCCGACGCATCCGATCCGT
 CTGGGTCTGGCCCTGAACCTCAGTGTTTTCTATTACGAAATCCTGAACAGTCCGGAAAAAGCGGCAT
 CCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGACACGCTGTCAGAAGAATCGTACAA
 AGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACCCTGTGGACGTCCGATACCCAAGGC
 GACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT**TAA**

>14-3-3ζ_C25A_C189A_S58E_(ζ_S58E)

ATGGGCAGCAGCCATCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
 GCCATATG**GAAAACCTGTATTTCCAGAGT**GTCGAC**ATG**GATAAAAACGAACTGGTGCAGAAAGCGAA
 ACTGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCAATGAAATCTGTGACCGAACAGGGC
 GCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCTACAAAAATGTGGTTGGTGCACGTC
 GCAGCGAGTGGCGTGTCGTGAGTTCCATTGAACAGAAAAACGGAAGGTGCTGAGAAAAAACAGCAAAT
 GGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGATATCTGTAACGACGTCCTGAGCCTG
 CTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTAAAGTTTTCTATCTGAAAATGAAAG
 GCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGACAAAAAAGGTATCGTCGATCAGAG
 CCAGCAAGCTTACCAAGAAGCGTTCGAAATCTCTAAAAAAGAAATGCAGCCGACGCATCCGATCCGT
 CTGGGTCTGGCCCTGAACCTCAGTGTTTTCTATTACGAAATCCTGAACAGTCCGGAAAAAGCGGCAT
 CCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGACACGCTGTCAGAAGAATCGTACAA
 AGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACCCTGTGGACGTCCGATACCCAAGGC
 GACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT**TAA**

>14-3-3ζ_C25A_C189A_S58E_Ntail_(ζ_S58E_Ntail)

ATGGGCAGCAGCCATCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCAGCCACAAGACTG
 GCCATATG**GAAAACCTGTATTTCCAGAGT**GTCGACGCGTGTAAGGTAGTAGCGGTGGT**ATG**GATAA
 AACGAACTGGTGCAGAAAGCGAACTGGCCGAACAAGCAGAACGTTATGATGACATGGCGGCCGCA
 ATGAAATCTGTGACCGAACAGGGCGCCGAACCTGTCAAACGAAGAACGTAATCTGCTGTGCGTTGCCT
 AAAAAATGTGGTTGGTGCACGTCGCAGCGAGTGCGTGCTGAGTTCCATTGAACAGAAAACGGA
 AGGTGCTGAGAAAAAACAGCAAATGGCACGTGAATATCGCGAAAAAATTGAAACCGAACTGCGTGAT
 ATCTGTAACGACGTCCTGAGCCTGCTGGAAAAATTTCTGATTCCGAATGCGAGCCAGGCGGAATCTA
 AAGTTTTCTATCTGAAAATGAAAGGCGATTACTACCGCTACCTGGCGGAAGTTGCAGCTGGCGATGA

CAAAAAAGGTATCGTCGATCAGAGCCAGCAAGCTTACCAAGAAGCGTTCGAAATCTCTAAAAAAGAA
ATGCAGCCGACGCATCCGATCCGTCTGGGTCTGGCCCTGAACTTCAGTGTTTTCTATTACGAAATCC
TGAACAGTCCGGAAAAAGCGGCATCCCTGGCCAAAACCGCATTTCGATGAAGCGATTGCCGAACCTGGA
CACGCTGTCAGAAGAATCGTACAAAGATAGTACCCTGATCATGCAGCTGCTGCGCGACAACCTGACC
CTGTGGACGTCCGATACCCAAGGCGACGAAGCAGAAGCTGGTGAAGGCGGTGAAAAT TAA